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Development of lower-risk guidelines for online sports and race betting in Australia using objective behavioural data

Robert M. Heirene^{1*}, Dilushi Chandrakumar¹, Gary Fahey¹, Eunice L. Y. Huynh¹ and Sally M. Gainsbury¹

Abstract

Background Lower risk gambling guidelines provide clear, actionable advice for reducing the risk associated with gambling by specifying behaviour associated with non-harmful use. Here we aimed to develop lower-risk guidelines specific to online sports and race betting in Australia that [1] captured behaviours unique to this activity associated with high risk of experiencing gambling harms (e.g., depositing frequency, number of sports & races bet on) and [2] were based on objective behavioural data from a gambling operator not subject to self-report inaccuracies.

Methods Customers from two large online wagering sites in Australia owned by the same parent company were invited to take part in a survey including the Problem Gambling Severity Index (PGSI) and the Gambling Harm Measure (GHM). Survey responses were matched to customer's account data provided by the operator, including all bets and transactions placed six months before and after the survey. The final sample consisted of 1,647 participants (84.6% male; mean age = 44.1).

Results Receiver Operating Characteristic curve analyses were used to identify optimal threshold values for PGSI and GHM harm separately across eight behavioural indicators: frequency, spend, percentage of household income spent, number of different sports/races bet on, deposit frequency, deposit amount, percentage of household income deposited, and number of active wagering accounts (self-reported). All indicators had acceptable Area Under the Curve values (AUC; ≥ 0.6) except gambling frequency. Optimal threshold values were higher for GHM than PGSI harm, and higher than those established in studies using self-report data (e.g., $\leq 6\%$ vs. $\leq 1\%$ of monthly income spent) irrespective of the measure of harm. A dose-response relationship was observed, whereby surpassing more limits was generally associated with greater odds of reporting harm.

Conclusions Our findings provide preliminary support for the following Lower-Risk Guidelines for Online Sports and Race Betting in Australia: [1] make no more than four deposits per month, [2] deposit no more than 2% of your pre-tax household income per month, [3] bet with no more than two different sites or apps, and [4] stick to betting on four or fewer different types of sports or races. We strongly encourage further evaluation and testing of these guidelines to confirm their value across samples and over time.

*Correspondence:
Robert M. Heirene
robert.heirene@sydney.edu.au

Full list of author information is available at the end of the article



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Keywords Low risk gambling guidelines, Gambling disorder, Problem gambling, Gambling-related harm, Harm reduction, Policy

Background

Originating in Canada, several research groups have attempted to establish ‘lower-risk’ guidelines for people who gamble [1, 2]. These guidelines assume a consumption or dose-response relationship between engagement with gambling (e.g., time and money spent) and the risk and/or harm relating to it [3]. The guidelines are analogous to those provided for lower-risk alcohol consumption (e.g., “Drink no more than 10 standard drinks in a week and no more than four standard drinks on any day”, 4). While the guidelines do not endorse any ‘safe’ level of gambling, they offer practical advice for people who choose to gamble on how to do so in a lower risk manner. In doing so, they may provide an effective alternative to either proposing abstinence or the ambiguous “gamble responsibly” narrative that has historically dominated public health messaging campaigns despite evidence of its inefficacy as a harm reduction strategy [5, 6].

The Canadian lower-risk guidelines [7] were derived from cross-provincial survey data and include the following recommendations: [1] gamble no more than 1% of household income (pre-tax), [2] gamble no more than 4 days per month, and [3] avoid engaging with two or more types of gambling in a month. Using population prevalence surveys from eight different countries, Hodgins et al. [8] found some variation in the most appropriate values for these limits, although they remained within a restricted band (e.g., gambling no more than 5 to 8 days per month, spending no more than 1 to 3% of gross household income). Other researchers have developed similar guidelines specific to jurisdictions [9, 10]. For example, the Australian population guidelines include gambling no more than 0.83 to 1.68% of personal income per year, spending no more than \$380–\$615 per year on gambling, gambling on no more than 20 to 30 days per year, and avoiding regularly gambling at more than two types of games [11].

Existing research has provided a solid foundation for understanding appropriate limits for lower-risk gambling behaviour. It is, however, limited by at least two significant factors. First, all but one European-based investigation [12] used self-reported gambling behaviour to develop lower-risk limits. That is, people were *asked* how much money they had spent and how many days they gambled over the past year with all types of gambling. These values were subsequently correlated with self-reported harm to identify appropriate values that appeared best to differentiate harmful and non-harmful gambling. This is problematic as research shows that most people are highly inaccurate in reporting their past

30 day behaviour on one gambling site [13, 14], let alone when asked to summarise past year behaviour for all types of gambling. In the study by Heirene et al. [14], the median difference between actual and estimated net outcome on an online wagering site in Australia was 100% of the actual value (e.g., someone who thought that they lost \$1,000 typically actually lost \$2,000, or \$0). Studies also find the accuracy of self-reporting declines with increased betting and expenditure, further complicating the relationship between self-reported spend and the experience of harm [13–15].

Second, research has almost exclusively focused on developing limits that are applicable to all forms of gambling (cf., [16]). This, as noted by Currie et al. [1], is problematic as different types of gambling (e.g., lottery, sports betting, electronic gaming machines [EGMs]) are associated with different levels of risk in terms of the associated likelihood of developing gambling problems and harms [17]. There are parallels here with alcohol consumption – different types of alcohol bring different risks for consumers and therefore the guidance for one type (e.g., spirits) should not be the same for others (e.g., beer, [18]) given differences in ethanol concentration (referred to as *alcohol by volume* [ABV] and often expressed as a percentage of total volume). However, while alcohol can be standardised to single units, thus permitting cross-type recommendations, this cannot be achieved for gambling where the behaviour varies from type to type (e.g., maximum bet size, speed of play, bet-outcome delay, and the expected rate of return). Few studies have attempted to develop limits for specific forms of gambling [16, 19], and even fewer have focused on developing guidelines that target key risk behaviours unique to the activity (e.g., depositing money into one’s account for online gambling, or time on device for EGM gambling). Variability between the limits identified for an online sample versus those established for land-based gambling led Louderback et al. [12] to conclude that unique limits might apply to different gambling populations. Consistent with this, 77% of gambling experts and 73% of the general public surveyed by Dowling et al. [20] believed lower-risk guidelines should be developed for each type of gambling.

Developing limits specific to online forms of gambling may be particularly beneficial given their rapid growth and unique risk factors. Online gambling is the fastest growing form of gambling in Australia and worldwide [21] and has been linked to elevated rates of problematic gambling [22, 23], particularly among those who gamble with their mobile device [24]. Unlike traditional land-based or retail gambling, online gambling is accessible

anytime and anywhere via mobile devices, with minimal physical or social barriers to potentially excessive spending [25, 26]. Whilst there is a need for broad regulatory measures that attempt to prevent online gambling harms at the population level [27], there is also a need to equip all those who gamble online with the knowledge of how to do so in a way that minimises their risk of experiencing harm. Identifying levels at which gambling is less likely to result in harms can also assist regulators and operators to identify those who may be gambling in a risky manner to guide proactive intervention.

Online betting is one of the most common forms of gambling engaged in by younger generations [21, 28], who are often the most at risk of experiencing gambling-related harms [29]. A recent systematic review of 21 studies from 18 countries found younger adults (from 18 up to 34 years) were 1.51 times more likely than middle-aged adults to report problematic gambling, although the criteria used to determine problem gambling status varied between studies [30]. Young people in Australia and internationally are now exposed to gambling products and marketing in a variety of novel ways, including through social media [31], affiliate marketing via so-called “influencers” [32], sponsoring of sports teams [33], and TV advertisements during and around events known to attract young people (e.g., sporting events, [34]). The elevated risk of gambling problems among young people [22, 35] has been recognised in recent government gambling policy in the Netherlands [36]. This policy requires a lower automatic deposit limit for those aged 18–24 (€300 a month) versus those aged 25 and over (€700 a month). Few studies have attempted to determine whether lower-risk guidelines should differ for younger populations, who are likely to have fewer resources to offset the negative impact of any of gambling they engage with. Currie et al. [1] reported that risk curves showing the relationship between behaviours (e.g., gambling spend) and harm rates appeared similar for younger and older populations after dividing their sample using a median split. However, the absence of tailored lower-risk guidelines for young adults—those most vulnerable to harm as they navigate their early exposure to legal gambling—represents a gap in the research on this topic.

Lower-risk gambling guidelines have strong potential to offer consumers clear, actionable advice they can follow to maintain lower risk levels of gambling. Thus, we aimed to develop lower-risk guidelines that are [1] specific to online sports and race betting in Australia¹ (the only legal form of online gambling other than lottery in Australia), [2] reflect the risk behaviours relevant to that particular type of gambling (e.g., amount of money deposited into one’s account), and [3] derived from a

dataset that combines objective behavioural data with self-reported problem gambling scores to ensure threshold values for the guidelines are not biased by inaccuracies in self-reported behaviour. To achieve these aims, we invited customers from two online sports and race wagering sites in Australia to take part in a survey about their gambling and linked their responses to the behavioural account data provided by the sites. The survey included the Problem Gambling Severity Index (PGSI; [37]) and the Gambling Harm Measure [38], allowing us to develop and compare lower-risk limits for two different measures of gambling harm.

Our hypotheses are specified in detail in our study pre-registration (<https://doi.org/10.17605/OSF.IO/BX43K>) and are summarised below:

H1 *We will be able to develop lower-risk limits for eight different behavioural indicators using actual betting data that can accurately classify harmed and non-harmed participants. Consistent with prior research and conventional guidelines for interpreting area under the curve (AUC) values, we will consider a limit to have acceptable classification accuracy if it has an AUC value equal to or greater than 0.60.*

H2 *Participants who surpass each of our newly derived lower-risk limits in the six months preceding survey participation will have greater odds of reporting the experience of harm compared to those who do not surpass the limits after accounting for the impact of demographic factors (i.e., age, gender, employment status, and education level) and other gambling (i.e., total number of wagering accounts used in the past 12-months, and any past-30-day spending on other forms of gambling) on harm. This will be the case when [H2.1] each of the eight limits are examined in separate statistical models and [H2.2] when all are included simultaneously.*

H3 *Participants who surpass successively more of the eight limits will have increasingly greater odds of reporting the experience of harm after accounting for the impact of demographic factors (i.e., age, gender, employment status, and education level) and other gambling (i.e., whether they engage with other forms of gambling outside of online wagering) on harm.*

H4 *Dowling et al. [19] provide some of the most comprehensive analysis regarding activity-specific limits. They used data from population-level surveys in two Australian states (collected 2011–2015) to develop limits for several types of gambling (EGMs, horse/dog racing, instant scratch tickets, lotteries, Keno, Casino table games, bingo, and sports/other event betting), although they noted the small cell sizes available for analyses relating to certain types of*

¹ Referred to mostly as simply “wagering” hereafter.

gambling. We predicted that the limits we developed would be better at differentiating between harmed and unharmed participants than the limits identified by Dowling et al. [19] for both horse/dog racing and sports/other events betting as these were derived from self-reported behaviour and not specific to online forms of betting.

H5 *Participants who report experiencing harm in the survey will be more likely than those who do not report experiencing harm to subsequently surpass our newly derived limits.*

Each hypothesis was, as per our pre-registration, assessed twice, first using the PGSI as the measure of harm and second using the GHM as the measure of harm. As an exploratory, post-preregistration aim, we sought to identify lower-risk limits specific to young people aged 18 to 25 by isolating this group in our sample and repeating the analyses used to test H1. We focused on participants aged 18 to 25 to be mostly comparable with the age range applied in the Netherlands' deposit limit policy (18 to 24) and consistent with the age bracket supported by *Headspace*, an Australian mental health service for young people (up to 25).

Methods

We preregistered our overall project plan for how we selected and recruited participants on Open Science Framework (OSF; <https://doi.org/10.17605/OSF.IO/VDSMW>) separately to the hypotheses and analyses reported here. We later preregistered our hypotheses and analysis plan for the study reported here after data collection but before analysis (<https://doi.org/10.17605/OSF.IO/BX43K>). Unless otherwise stated, we followed all methods as outlined in our pre-registrations.

Participants & recruitment

Customers from two licensed online wagering platforms in Australia, operated by the same international parent company, were invited to participate in a survey about their gambling behaviour. To be eligible, customers needed to have placed at least one bet within the preceding six months, held an account for a minimum of 30 days, and not have a suspended account, be self-excluded, or be on a timeout/take-a-break.

Once eligibility was established, two groups were identified. First, the operator flagged all eligible customers who had been classified as "at-risk" according to their risk detection system within the prior six months (this allowed us to separately evaluate this system). Second, a random sample of "not-at-risk" customers was drawn at a 20:80 ratio of at-risk to not-at-risk customers. This resulted in invitations being sent to 4,829 "at-risk" and 20,000 "not-at-risk" customers, rounding to the nearest thousand. Our preregistration specified a target of

at least 1,500 completed surveys, defined as those where all Problem Gambling Severity Index (PGSI) items were answered.

Of the 24,829 individuals invited, 3,867 accessed the survey, 1,959 completed up to the PGSI, and 1,470 provided responses to all questions. Separate analyses indicated that the survey sample (i.e., those completing at least the PGSI items) was largely representative of the invited population in terms of demographic and gambling characteristics [39], though the sample tended to be slightly older and included more frequent and recent bettors. Compared to non-responders, survey responders bet more frequently over the past six months (Cohen's $d = 0.58$), had placed a bet more recently ($d = -0.51$), were older ($d = 0.40$), and made more deposits ($d = 0.32$).

Procedures

In January 2024, we received a dataset containing unique customer identification numbers for all 24,829 eligible participants. Unique invitation links were generated through Qualtrics, where the survey was hosted, and matched to each identifier. These links were provided to the operator, who disseminated them via an initial email invitation, followed by an SMS reminder eight days later. All participants reviewed an information sheet and provided informed consent before starting the survey. Upon completion, participants were given the option to enter a prize draw for one of 20 e-gift vouchers worth \$250, which could not be redeemed for cash or used for gambling. Ethical approval for the study was granted by the University of Sydney Human Ethics Committee [ID: 2023/029]. Our overall project preregistration outlines further methodological details, including participant recruitment, survey measures, the invitation email, and the account data collected (<https://osf.io/vdsmw>).

Measures

Behavioural account data

The online wagering operator supplied behavioural account data for all 24,829 customers invited to the survey, covering a period of six months prior to survey launch and six months following its close. This dataset included detailed records of all bets and financial transactions (i.e., deposits and withdrawals) made during that 12-month period. Demographic information, such as age, sex, postcode, and account registration date, was also provided. To better understand long-term engagement with consumer protection tools, data on the use of features like deposit limits, timeouts/take-a-breaks, and self-exclusion were provided for the two years leading up to the survey invitation.

Outcome variables

PGSI harm index: The PGSI is a commonly used problem gambling screening measure with nine items scored on a four-point Likert scale from “Never” to “Always” (e.g., “How often have you bet more than you could really afford to lose?”; [37]). Items in our survey focused on experiences over the past 12 months. Consistent with previous investigations of lower-risk guidelines [1, 11], we focused on the seven items relating to gambling harms within the PGSI (i.e., feeling guilty, losing more than one can afford, recognition of the problem, health problems, financial problems, criticism from others, and borrowing money). The two items that relate to needing to gamble with more money and loss chasing (typically ordered as items 2 and 3) were omitted as these focus on behaviours that are indicative of problematic or disordered gambling, but do not necessarily represent experienced harm. For each of the seven harm-focused items, participants were assigned a score of 0 (representing a response of “Never”) or 1 (representing a score of “sometimes”, “most of the time”, or “almost always”). A score of 2 or more across these seven items was used to classify a participant as harmed, again consistent with prior research using this same index [1, 20].

GHM harm index: For our second harm index, we used the Gambling Harm Measure [38], a 16-item scale measuring the impact of gambling across six life domains (financial, psychological, relationship, physical health, work/study, and legal). Gooding et al. [38] share the full list of GHM items in the recent evaluation of the measure, which found strong correlations between the GHM and other measures of gambling-related harm. Dichotomous responses (yes/no) are provided to all questions which are asked in relation to the last 12 months (e.g., In the last 12 MONTHS, have you experienced any serious relationship consequences because of your gambling? *For example, have you lost friends or family; or experienced separation or divorce; or engaged in physically violent arguments?*). Questions are asked in a graded manner with three questions for each domain, the first asking about prioritising gambling over an activity/responsibility, the second asking about strains or pressures caused by gambling in that domain, and the third asking about serious consequences of gambling in that domain (the third level of questioning is only presented to participants if they say they have experienced harm at the strain/pressure level in that domain). Any participant who responded “yes” to a question at the second level of any domain (i.e., strain/pressure level of harm) was classed as having experienced harms. Note that the legal domain only has one question (“In the last 12 MONTHS, have you done anything illegal due to your gambling? For example, have you stolen money or valuables, or committed fraud or embezzlement, etc.”). A positive response

to this question also resulted in a participant being classed as harmed.

Participants’ scores on the PGSI and GHM were strongly and positively correlated (weighted Spearman’s rank test: $r_s [1,578] = 0.71$).

Behavioural indicators used for lower-risk guidelines

Behaviours used to derive lower-risk guidelines were generated using a combination of behavioural account data and survey responses. We first generated the following four behavioural indicators that were analogous to those studied in other lower-risk gambling guideline studies (i.e., gambling frequency, spend, spend as a percentage of income, & breadth of gambling activity):

Gambling frequency per month: The average number of active betting days (i.e., days in which at least one bet was placed) per month in the six months preceding participation.

Monthly spend on bets: The average amount staked/spent on bets per month in the six months preceding participation. Note, this refers only to the amount bet, and not the amount lost.

Percentage of monthly household income spent: In the survey, participants were asked to indicate which household income bracket most closely reflected their annual income (e.g., \$50,000 - \$59,999 per year, \$60,000 - \$79,999 per year) and as such, we did not have exact income values. Therefore, to estimate annual income, we first took the middle/midpoint value within each bracket to represent each person’s annual income (e.g., for the bracket “\$80,000 - \$99,999/yr”, the middle value would be \$90,000). However, one of the response options was an income of more than \$200,000 a year. To determine an appropriate value that would split this population equally, we used ABS census data which contains higher income brackets than we used. Using this data, we computed the total number of people in income brackets above \$200,000 a year and identified the midpoint value within these brackets that most closely split those earning more than \$200,000 into two even groups: \$257,374. Estimated annual income values for all participants were divided by 12 to estimate monthly income. Next, participants’ monthly spend on bets value (as created above) was divided by this monthly income value and multiplied by 100 to calculate the average percentage of monthly household income spent each month. Only 1,055 participants reported their income (responses of “Unknown” and “I’d rather not say” were permissible).

Number of different activities (i.e., number different of sports/races bet on): The total number of different sports and/or races bet on in the six months preceding participation. We used the broadest category of activity type available in the data, which groups by the name of the

overall sport/event type (e.g., tennis, boxing, horseracing). There were at least 33 different activity types present in the data, with horseracing, greyhound racing, harness racing, basketball, and soccer being the five most common activities by total bets placed in the month preceding the survey.

Given the specific features and nature of online wagering, we also looked at the following activity-specific indicators which have not (to our knowledge) been examined in previous studies:

Total monthly number of deposits: The average number of deposits made per month in the six months preceding participation.

Monthly deposit amount: The average amount of money deposited into a participant's account per month in the six months preceding participation.

Percentage of monthly household income deposited: This variable was created using a similar process to that described for the Percentage of monthly household income spent variable. That is, we first calculated participants estimated monthly income, divided their average monthly deposit amount by this value, and multiplied this by 100 to compute the average percentage of monthly household income deposited per month. We restricted this value to a maximum of 100 to be consistent with prior research on this subject [12].

Number of active wagering accounts: All participants in the survey were asked: "Including this account, how many online gambling accounts have you used in the past 12 months?" Response options were: 1, 2, 3, 4, and 5+ (converted to a continuous variable, with 5+ changed to "5").

Variations of all behavioural indicator variables were also created to represent behaviour over the six months following the survey to explore post-survey breaches of identified limits. Whilst other studies have looked at other indicators, including spend variability and slope of spend over time [12], we focused on the above four behaviours (beyond the standard four used in previous studies) as we believe they are easily understood by consumers and therefore any resulting lower-risk guidelines proposed will be actionable with little explanation.

Demographic and other gambling control variables

Several additional variables were used as covariates in our regression analyses, as per our preregistration. These included age and gender from account data, and employment status (unemployed vs. employed), education level, and other recent gambling (i.e., a dichotomous variable indicating the presence or absence past-30-day spending on other forms of gambling, such as Lottery, EGMs, Casino table games etc.) from survey responses. Our preregistration further outlines how these variables were

collected and operationalised in our analyses (<https://doi.org/10.17605/OSF.IO/BX43K>).

Data analysis

All statistical analyses were conducted using R version 4.4.2 [40] and followed the five steps outlined in our preregistration. The assumptions underlying all statistical models—including multicollinearity, normality of residuals, and the impact of influential observations—were evaluated using the 'check_model()' function from the 'performance' R package [41]. Our Analysis Document presents our analysis code alongside all outcomes presented here and supplementary findings, including full statistical model outputs and assumption testing outcomes (<https://robheirene.quarto.pub/lower-risk-guidelines-for-online-wagering-analysis-document/>). We used the conventional alpha level of $p < 0.05$ to determine statistical significance for inferential tests.

Stage I: We first determined whether each participant met the criteria for harm using the PGSI and GHM indices and computed unweighted frequency counts and weighted percentages for participants in harmed and unharmed categories. Weighted percentages accounted for our oversampling of "at-risk" customers flagged by the operator's risk detection system by applying sample weights (adjusted for response rates) that adjusted for the relative underrepresentation of "not at-risk" customers.

Stage II (H1): Optimal cut-off points for each of the eight behavioural indicators of risk were identified using three different approaches. For all approaches, we used 'cutpointr' [42], an R package designed specifically to assist with identifying optimal limit values in binary classification scenarios. The three approaches to identifying cut-offs were [1] balancing specificity and sensitivity using the Youden index, [2] maximising sensitivity whilst ensuring a minimum specificity of 0.5, and [3] maximising specificity whilst ensuring a minimal sensitivity of 0.5. We used bootstrapping with replacement and 2,000 samples to ensure the optimal cut-off identified were robust to variations in sampling procedures and produce 95% confidence intervals (CIs) for Area Under the Receiver Operating Characteristic curve values (hereafter "AUC" values). AUC values reflect a behavioural indicator's overall ability to discriminate between harmed and unharmed groups independent of any specific cut-off point. We used conventional guidelines (e.g., 43,44) to interpret AUC values as failed (0.50–0.59), fair/acceptable (0.60–0.79), good (0.80–0.89), or excellent (0.9–1.0). Per our preregistration, we considered a behavioural indicator to have acceptable classification ability if it had an AUC value of 0.6 or more. It is important to note that some studies use 0.70 as a cut-off for acceptable AUC values, although this varies [44] and we have selected 0.60 as our minimum acceptable level as the most similar study

to ours that used actual betting account data to develop lower-risk guidelines found most AUC values in the range of 0.497 to 0.664 [12].

Sensitivity, specificity, and accuracy values were computed for all cut-offs identified. Of the three limit values returned for each behavioural indicator (Youden index, sensitivity, & specificity maximised), the one with the highest accuracy rating (i.e., total proportion of cases correctly classified) was selected for further examination.

Stage III (H2-3): Next, we examined the ability of surpassing the new limits to predict both harm indices after adjusting for participants' demographic characteristics (i.e., age, gender, employment status, and education level) and other gambling behaviour (i.e., number of additional online accounts² and past-30-day spending on other forms of gambling). Logistic regression models with weighting to account for the overrepresentation of "at-risk" customers were used to separately test the impact of passing each limit in the six months preceding survey participation (defined as having a monthly average value above the cut-off) on both the PGSI and GHM harm indices. Additional logistic regression models were performed that included demographic and other gambling covariates and all limits simultaneously to determine their unique impact on [1] PGSI harm and [2] GHM harm. Lastly, we performed two further logistic regression models (one each for the PGSI & GHM harm indices) that used the total number of limits a person surpassed (0–8) as a predictor alongside the same demographic and gambling covariates.

Stage IV (H4): The 'caret' R package was used to compute performance metrics for Dowling et al.'s [19] limits and compared these to those computed via 'cutpointr' for our newly developed limits. We extracted the limits from Dowling et al. [11] most relevant to our sample of online sports and race wagerers for comparison: sport and other events betting and horse/dog racing. We focused only on the three behavioural indicators examined in our study and by Dowling et al. [19]: gambling frequency per month, monthly spend on bets, and percentage of monthly household income spent on gambling. Per our pre-registration, we repeated the

regression analyses conducted in Stage III for Dowling et al.'s limits to determine their ability to predict harm (model outputs from this process are presented as supplemental outcomes in our Analysis Document).

Stage V (H5): Lastly, we computed the number and percentage of participants in each harm status group (according to PGSI & GHM) whose average monthly gambling surpassed each threshold in the six months before and after survey participation³.

Results

The 1,647 people who completed all PGSI items formed our initial sample for analyses. Of this sample, 84.6% were male (1.6% were recorded as having an "unknown" gender by the operator) and the mean age was 44.1 years ($SD = 15.5$)⁴.

Rates of harm

Among the 1,647 people who completed the PGSI items, 893 met the criteria for experiencing harm (equivalent to 50% of the weighted sample). Among the 1,580 who completed all items of the GHM, 472 (25% of the weighted sample) met the criteria for experiencing harm. Almost all those meeting criteria for harm according to the GHM met PGSI criteria for harm; however, approximately half meeting PGSI criteria did not meet GHM criteria, suggesting a higher threshold for positive classification (i.e., experience of harm) using the GHM (see Table 1). Figure 1 illustrates this nesting of harm groups, displaying the unweighted counts for harm groups in the form of a Venn diagram.

Determining optimal cut-offs for lower-risk guidelines

The optimal thresholds identified for each behavioural indicator are displayed in Table 2. Largely supporting H1, all behavioural indicators had acceptable AUC levels for both harm indices other than betting days per month, which had AUC values below 0.6 for both PGSI and GHM harm. AUC values were typically larger for the online wagering-specific behavioural indicators (i.e., number of deposits, amount deposited, percentage of income deposited, & number of active online gambling account) compared to those explored in previous studies (i.e., betting days, spend, percentage of income spent, & number of different activities), averaging 0.65 for the former and 0.61 for the latter for PGSI harm and 0.66 and 0.63 for GHM harm, respectively. AUC values were

² This is one of the behavioural indicators for which we developed a low-risk guideline and so a participant's number of active account was not included as a covariate in the model testing the impact of surpassing the cut-off for this indicator as identified in Stage II.

Table 1 Rates of harm for GHM grouped by PGSI status

	PGSI: harm (%)	PGSI: No harm (%)
GHM		
Harm	448 (47.56)	24 (2.71)
No harm	406 (52.44)	702 (97.29)

values = n (unweighted) (%), weighted); PGSI, problem gambling severity index; GHM, Gambling harm measure

³ In our preregistration, we stated that we will only compute these rates for the six months after participation, but we have reported rates of passing limits before and after the survey for a more comprehensive understanding. Weighted percentages are presented in our Analysis Document, Supplemental Table 2.

⁴ All values in this sentence are weighted to account for the oversampling of customers flagged by the operator's risk detection system.

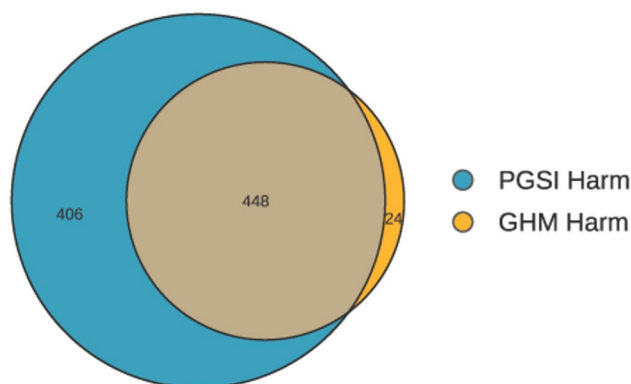


Fig. 1 Venn diagram: Overlap in PGSI and GHM harm groups. This Venn diagram shows overlap in participants meeting criteria for either PGSI harm, GHM harm, or both. Values represent the unweighted count of participants in each harm grouping. PGSI, problem gambling severity index; GHM, gambling harm measure

typically higher for GHM harm classification than PGSI harm.

Optimal limits (i.e., those that maximised accuracy) varied by behavioural indicator and harm index. Maximising Youden and sensitivity values led to cut-offs with the highest accuracy levels for PGSI harm, whereas maximising Youden and specificity values led to the most accurate cut-offs for GHM harm. The three cut-off points identified for all behavioural indicators using the different approaches (Youden, sensitivity & specificity) and their corresponding performance metrics are presented in our Supplemental Table 1 of Analysis Document⁵.

For PGSI harm classification, the following limits led to the highest sensitivity values, suggesting larger proportions of harmed participants surpassed them: spending ~\$251 or more per month (sensitivity=0.69), having three or more active online wagering accounts (0.61), and depositing 2.13% or more of one's monthly household income per month (0.60). For GHM harm classification, the limits with the highest sensitivity values were

⁵ Direct link to table: <https://robheirene.quarto.pub/lower-risk-guidelines-for-online-wagering-analysis-document/#table-all-limit-cut-offs>.

Table 2 Optimal limit values for lower-risk guidelines for online wagering with classification performance metrics

Behavioural indicator	PGSI harm			GHM harm						
	AUC [95% CIs]	Metric maximised	Optimal cut point (≥)	Accuracy	Sensitivity/ Specificity	AUC [95% CIs]	Metric maximised	Optimal cut point (≥)	Accuracy	Sensitivity/ Specificity
Analogues of standard behavioural indicators used in lower-risk gambling guideline studies										
Betting days per month	0.560 [0.532, 0.587]	Sensitivity	5.68	0.546	0.579/0.507	0.573 [0.544, 0.603]	Specificity	8.56	0.573	0.498/0.605
Spend per month	0.621 [0.594, 0.648]	Sensitivity	\$251.42	0.599	0.685/0.497	0.648 [0.619, 0.678]	Specificity	\$2108.33	0.665	0.494/0.737
% of monthly income spent per month	0.655 [0.622, 0.688]	Youden	6.38%	0.627	0.580/0.686	0.667 [0.631, 0.703]	Youden	8.92%	0.674	0.635/0.692
Number of activities	0.611 [0.583, 0.637]	Youden	4.08	0.568	0.499/0.650	0.618 [0.588, 0.647]	Youden	4.58	0.601	0.551/0.623
Online wagering specific behavioural indicators not previously examined										
Number of deposits per month	0.651 [0.624, 0.676]	Youden	3.81	0.626	0.623/0.630	0.669 [0.642, 0.699]	Specificity	14.01	0.677	0.492/0.756
Amount deposited per month	0.648 [0.621, 0.674]	Sensitivity	\$80.17	0.619	0.717/0.504	0.667 [0.637, 0.696]	Specificity	\$904.50	0.668	0.494/0.742
% of monthly income deposited per month	0.691 [0.66, 0.724]	Youden	2.13%	0.651	0.604/0.709	0.690 [0.656, 0.723]	Specificity	9.62%	0.695	0.496/0.788
Number of active online gambling accounts	0.610 [0.585, 0.636]	Youden	3.00	0.592	0.606/0.576	0.612 [0.581, 0.64]	Youden	3.24	0.647	0.375/0.764

PGSI= problem gambling severity Index; GHM = gambling harm measure; AUC = area under the curve; CIs = confidence interval. Metric maximised = the approach used to reach the limit with the highest accuracy value for each behavioural indicator

spending 8.92% or more of one's monthly household income per month (sensitivity = 0.64), betting on 4.58 or more different activities (0.55), and betting on 8.56 days or more per month (0.50).

The following limits had the highest specificity values for PGSI harm classification, indicating that fewer non-harmed participants surpassed them: depositing 2.13% or more of one's monthly household income per month (specificity = 0.71), spending 6.38% or more of one's monthly household income per month (0.69), and betting on 4.1 or more different activities (0.65). For GHM harm classification, the limits with the highest specificity values were depositing 9.62% or more of one's monthly household income per month (specificity = 0.79), having 3.24 or more active online wagering accounts (0.76), and making 14 or more deposits per month (0.76).

Predictive ability of the optimal lower-risk guidelines

The optimal cut-off points identified for behavioural indicators in the prior section were, where appropriate, rounded to more easily interpreted and actionable values before including them in analyses presented in this section (e.g., the cut-off for number of deposits per month used to classify PGSI harm was rounded up from 3.81 to 4). Note that being *above*, rather than *at or above* the cut-offs was used to denote higher-risk gambling to align with the practical application of lower-risk guidelines (i.e., to suggest upper limits for behaviour rather than points at which behaviour becomes harmful).

The upper section of Table 3 displays outcomes from 16 binary logistic regression models (eight when PGSI harm was the outcome and eight when GHM harm was the outcome) testing H2.1—that is, individually examining the impact of surpassing each limit on harm (all 16 models included the demographic and other gambling covariates specified in our data analysis section [Stage III], though these are not presented in Table 3 for ease of interpretation). Supporting H 2.1, after accounting for the impact of demographic factors and other gambling, gambling above each was associated with significantly greater odds of harm on both the PGSI and GHM. The limit associated with the largest odds of reporting harm was depositing more than 2% of one's monthly household income per month for PGSI harm (odds ratio [OR] = 3.78, 95% CIs [2.70, 5.30]) and depositing more than 10% of one's monthly household income per month for GHM harm (OR = 7.10, 95% CI [4.50, 11.2]).

The lower section of Table 3 presents the outcomes from two models (one each for PGSI and GHM) testing H2.2. These included covariates and all limits simultaneously to account for their individual impact on harm. Contrary to H2.2, the only limits that remained significantly predictive of PGSI harm were three online wagering specific behavioural indicators—depositing more

than four times per month, depositing more than 2% of one's monthly income per month, and having more than two active wagering accounts. For GHM harm, spending more than 9% of one's monthly household income per month, depositing more than 14 times a month, depositing more than 10% of one's household income per month, and having more than three active wagering accounts remained uniquely predictive of harm. Being younger, male, and having recently spent money on other forms of gambling were significantly predictive of PGSI and GHM harm. Unexpectedly, betting more than six (PGSI harm) or 9 days (GHM harm) per month was associated with a reduced likelihood of experiencing harm in these limit-adjusted models.

Figure 2. presents the odds ratios and their 95% CIs from two final binary logistic regression models examining the relationship between the number of limits a participant passed in the six months preceding the survey and their harm status. For both PGSI and GHM harm, there was significantly increased odds of reporting harm among participants gambling above one and three or more (but not two) of the eight limits, with an almost monotonic rise in the odds of harm with each additional limit surpassed. Removing the betting frequency limit from these analyses pushed the lower CI bound for passing two limits closer to 1 for PGSI limits and above 1 for GHM limits, suggesting the absence of significantly increased odds of harm at two limits is at least partially explained by the inclusion of this limit (see subsection titled “*No. limits sans frequency*” of our Analysis Document for full outcomes from this exploratory analysis). Participants who gambled above all PGSI limits had 8.04 (95% CI [3.96, 16.53]) times greater odds of reporting harm than participants gambling below them. Participants who reported surpassing all GHM limits had 8.92 (95% CI [2.52, 31.65]) times greater odds of reporting harm. These findings are largely consistent with our H3 prediction.

Comparing new optimal lower-risk guidelines with previously established limits

Table 4 compares the performance metrics (accuracy, sensitivity & specificity) of the optimal limits identified here with those previously established for horse/dog race betting and *sports and other events betting* in Australia by Dowling et al. [19]. Supporting H4, the accuracy rate for all newly identified limits was higher than that for the previously established limits other than betting days per month for PGSI harm, which was marginally lower in comparison. However, our specific preregistered hypothesis for H4 was that the accuracy of our newly established limits would be at least 5% greater than that for Dowling et al.'s limits. This was the case for all limits when predicting GHM harm (accuracy was higher for our limits

Table 3 New optimal limits for online wagering as a predictor of harm status: demographic and other gambling adjusted logistic regression models

	PGSI Harm			GHM harm		
	OR	95% CI	p-value	OR	95% CI	p-value
Impact of isolated limits (covariate terms not shown; 16 models)						
<i>PGSI-specific limits</i>				<i>GHM-specific limits</i>		
> 6 betting days per month	1.37	1.07, 1.74	0.012	> 9 betting days per month	1.37	1.01, 1.85
> \$250 spent per month	2.00	1.56, 2.55	< 0.001	> \$2,100 spent per month	3.26	2.25, 4.74
> 6% of income spent per month	2.83	2.05, 3.93	< 0.001	> 9% of income spent per month	4.49	3.11, 6.49
> 4 different activities bet on	1.55	1.20, 2.00	< 0.001	> 5 different activities bet on	1.30	0.95, 1.77
> 4 deposits per month	2.60	2.01, 3.36	< 0.001	> 14 deposits per month	3.39	2.42, 4.75
> \$80 deposited per month	2.27	1.78, 2.89	< 0.001	> \$900 deposited per month	3.46	2.39, 5.01
> 2% of income deposited per month	3.78	2.70, 5.30	< 0.001	> 10% of income deposited per month	7.10	4.50, 11.2
> 2 active wagering accounts	2.17	1.71, 2.76	< 0.001	> 3 active wagering accounts	1.83	1.38, 2.44
<i>Impact of all limits (covariate terms shown; 2 models)</i>						
Age	0.98	0.96, 0.99	< 0.001		0.97	0.96, 0.99
Gender: Male	1.65	1.01, 2.71	0.047		2.64	1.32, 5.26
Gender: Unknown	3.01	0.83, 10.9	0.094		2.18	0.45, 10.7
Employment status: Unemployed	1.16	0.55, 2.43	0.7		1.58	0.72, 3.44
Educational attainment	0.93	0.85, 1.02	0.12		0.99	0.90, 1.10
Recent other gambling: True	2.22	1.46, 3.36	< 0.001		2.11	1.29, 3.45
<i>PGSI-specific limits</i>				<i>GHM-specific limits</i>		
> 6 betting days per month	0.36	0.22, 0.60	< 0.001	> 9 betting days per month	0.27	0.14, 0.51
> \$250 spent per month	1.18	0.68, 2.04	0.6	> \$2,100 spent per month	1.31	0.57, 3.01
> 6% of income spent per month	1.23	0.68, 2.22	0.5	> 9% of income spent per month	3.50	1.94, 6.34
> 4 different activities bet on	1.25	0.87, 1.81	0.2	> 5 different activities bet on	0.92	0.59, 1.43
> 4 deposits per month	2.25	1.36, 3.73	0.002	> 14 deposits per month	3.12	1.63, 5.98
> \$80 deposited per month	1.48	0.91, 2.40	0.12	> \$900 deposited per month	0.63	0.25, 1.55
> 2% of income deposited per month	2.21	1.25, 3.90	0.007	> 10% of income deposited per month	3.22	1.48, 6.98
> 2 active wagering accounts	1.94	1.41, 2.65	< 0.001	> 3 active wagering accounts	2.04	1.42, 2.95
<i>AIC/ BIC; No. Obs.; Nagelkerke's R Squared</i>						
1,317/ 1,385; 1,055*; 27.80%				1,100/ 1,168; 1,055*; 27.25%		

*Sample size reduced by 592 people who did not report income to determine % spent/deposited. All models not using % income variables used the full eligible sample (N=1,647). OR=odds ratio, CI=confidence interval. Reference for categorical predictors: Gender=female; Employment=employed; Recent other gambling=false

by 17.6–28.5%), but our PGSI-specific limits only outperformed Dowling et al.'s limits by a maximum of 2.9%.

Rates of gambling above new limits before and after the survey

Table 5 presents the rates of gambling above each of the new PGSI and GHM limits across all participants and when grouped by PGSI and GHM harm status, both before and after the survey. Rates refer to those whose average gambling across each six-month period surpassed the relevant thresholds. Participants reporting harm on either measure were more likely to surpass PGSI- and GHM-derived thresholds for every behavioural indicator both before and after survey participation, supporting H5.

Exploratory analyses

The following non-preregistered, exploratory analyses were conducted after accessing and analysing the data for this study. Consistent with Louderback et al. [12] and Currie et al. [1], we developed risk curves showing the relationship between behavioural indicators and the proportion of customers reporting harm (see Fig. 3). Our risk curves differ from Currie et al. who used survey response options as their x-axis values (e.g., gambling days per month of < 1, 1, 2, & so on) and Louderback et al. who used quintiles for this purpose. We instead chose to use deciles along the x-axes to provide a more granular visualisation of relationships. The plot for number of betting days shows the flattest line, indicating that this behaviour discriminates poorly between harmed and unharmed participants as increasingly higher numbers of

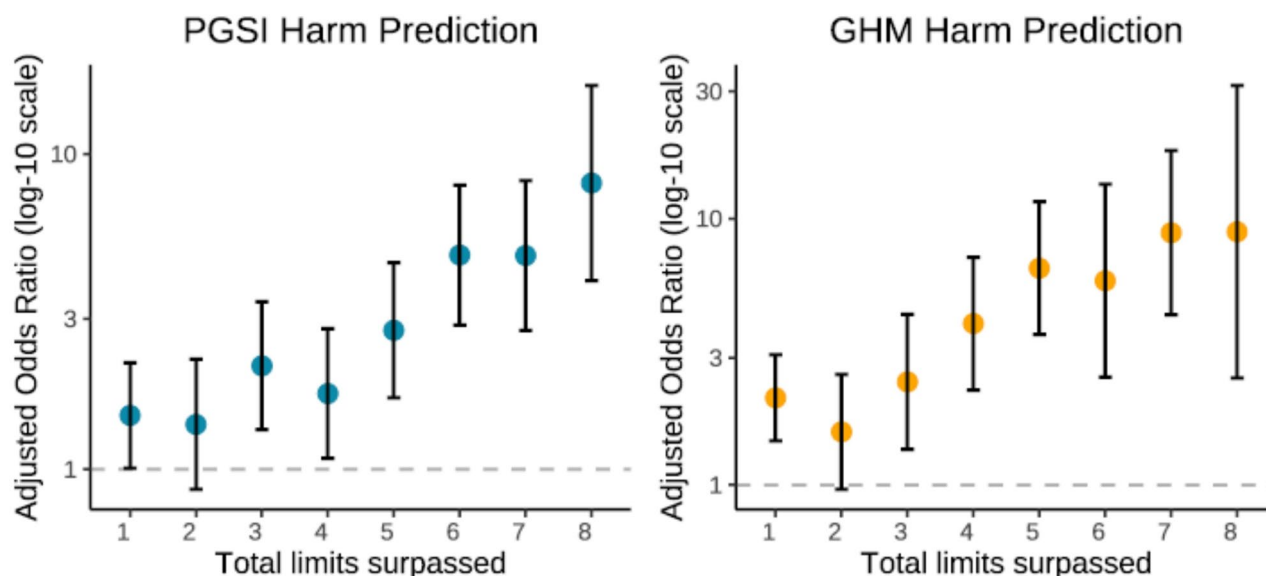


Fig. 2 Total number of limits surpassed as a predictor of harm. Outcomes from two logistic regression models examining the impact of passing successively more limits in the six months preceding survey participation on PGSI harm (left panel) and GHM harm (right panel). Dots represent odds ratios and error bars represent 95% confidence intervals (CIs). Both models included the following covariates: participant demographic characteristics (age, gender, employment status, & education level) and a dichotomous variable representing whether someone had spent money on other forms of gambling in the past 30 days. Y-axis are shown on a log-scale to enhance visualisation of 95% CIs for smaller odds ratios

betting days do not correspond to increases in the proportion of harmed participants. Behaviours that discriminate well between groups should be reflected by plot lines that start (left) low down the x-axis and rise sharply up rightwards, which here is best illustrated in the plot for percentage of monthly income deposited.

To determine whether limits were sensitive to age, we isolated all participants aged 25 or younger and repeated all analyses from Stage II to identify optimal threshold values for this subsample. Optimal cut-off points for all indicators identified for this subsample and their performance metrics are displayed in Supplemental Tables 3 and 4 of our Analysis Document⁶ and a summary of relevant findings is presented here. Harm rates were higher in this younger group of 239 participants, with 159 (67% [unweighted]) reporting PGSI harm and 87 (41%) reporting GHM harm. The optimal limit values for all behavioural indicators used to classify PGSI and GHM harm were lower for this subsample other than the number of wagering accounts, which had an identical (PGSI) or slightly higher value (GHM). The classification performance of all behavioural indicators was lower in this sub-sample than in the full sample. For PGSI harm, the only indicators with acceptable AUC values (i.e., ≥ 0.6) were percentage of monthly income spent (AUC=0.61, 95% CI [0.51, 0.71], optimal cut-off = 1.75%), percentage of monthly income deposited (AUC=0.62, 95% CI [0.52,

0.73], optimal cut-off = 0.74%), and number of active wagering accounts (AUC=0.61, 95% CI [0.51, 0.71], optimal cut-off = 3). For GHM harm, acceptable AUC values were found for frequency of deposits (AUC=0.62, 95% CI [0.54, 0.69], optimal cut-off = 9.49), amount deposited (AUC=0.61, 95% CI [0.53, 0.68], optimal cut-off = \$400.52), and number of active wagering accounts (AUC=0.61, 95% CI [0.51, 0.71], optimal cut-off = 3.7).

Finally, we later propose a set of recommended lower risk guidelines for sports and race betting based on our thresholds for PGSI-harm. These focus on percentage of income deposited, number of active wagering accounts, number of deposits, and number of different sports and races bet on. As we explain, these were selected based on a combination of statistical outcomes and practical considerations. An anonymous peer reviewer posed thoughtful questions about our choice of limits: Do they cover the largest number of those harmed? And could there be a cluster of people experiencing harm that are not covered by our limits (is there a “blind spot”)? It is difficult to compare every intersection of the eight limits (i.e., all possible combinations of passing or not passing each limit) to identify the most “statistically optimal” selection, given very low to no participant numbers in some intersections. However, we have performed additional, exploratory analyses which appear to support our choice of limits.

First, Fig. 4 shows all limit setting intersections represented by at least five participants. Fewer than eight participants were represented in intersections that did not

⁶ Direct link to tables: <https://robheirene.quarto.pub/lower-risk-guidelines-f-or-online-wagering-analysis-document/#table-all-limit-cut-offs-1>.

Table 4 Comparing the classification performance of new optimal limits vs. Dowling et al.'s [19] limits for race and sports betting

Limit	PGSI harm		GHM harm	
	Accuracy	Sensitivity/ specificity	Accuracy	Sensitivity/ specificity
Betting days per month				
Dowling et al. [19]: No more than 1.25 days (horse/dog racing)	0.550	0.840/ 0.206	0.398	0.867/ 0.198
Dowling et al. [19]: No more than 1.17 days (sport and other events betting)	0.549	0.840/ 0.204	0.397	0.867/ 0.197
New optimal limit PGSI: No more than 5.68 days New optimal limit GHM: No more than 8.56 days	0.546	0.579/ 0.507	0.573	0.498/ 0.605
Spend per month				
Dowling et al. [19]: No more than \$24 (horse/dog racing)	0.570	0.903/ 0.175	0.383	0.924/ 0.153
Dowling et al. [19]: No more than \$33.3 (sport and other events betting)	0.576	0.887/ 0.207	0.400	0.913/ 0.181
New optimal limit PGSI: No more than \$251.42 New optimal limit GHM: No more than \$2,108.33	0.599	0.685/ 0.497	0.665	0.494/ 0.737
% of monthly income spent per month				
Dowling et al. [19]: No more than 0.55% (horse/dog racing & sport and other events betting)	0.608	0.854/ 0.306	0.465	0.887/ 0.267
New optimal limit PGSI: No more than 6.38% New optimal limit GHM: No more than 8.92%	0.627	0.580/ 0.686	0.674	0.635/ 0.692

All limits previously established by Dowling et al. [19] were annual and have been converted to monthly values here by dividing by 12

include one or more of our limits, other than not surpassing any limit at all, which was the most prevalent intersection. PGSI rates for each intersection are presented in the form of box plots, and these do not reveal a missed group with high PGSI scores in any of the intersections that do not include one of our four limits.

Second, we calculated that only 8.4% of all participants meeting criteria for PGSI-harm did not pass one or more

of our limits in the preceding six months. Overall, 91.6% of harmed participants passed at least one of our four proposed limits, with 20.3% passing one, 19.2% passing two, 30.1% passing three, and 21.3% passing four.

Discussion

The aim of this study was to develop lower-risk guidelines specific to online wagering in Australia using survey responses with matched objective account data. We provide a discussion of our findings, their limitations, and practical implications below.

Technical identification of optimal lower-risk limits for online wagering

As per our preregistration, we selected the limit from three approaches (maximising Youden, sensitivity or specificity) that resulted in the highest overall accuracy. Maximising Youden and sensitivity values led to cut-offs with the highest accuracy levels for PGSI harm, whereas maximising specificity and Youden values led to the most accurate cut-offs for GHM harm. This likely results from base rate differences in harm between the two outcomes in our sample. As GHM-determined harm was only prevalent in just under 30% of the unweighted sample, increasing the limits and thereby not misclassifying non-harmed participants as harmed (i.e., increasing the true negative rate) led to the greatest increase in accuracy in GHM classification. By contrast, PGSI harm was more common, reported by slightly more than half of the unweighted sample; in this scenario, overemphasising either sensitivity or specificity will have a less biased impact on accuracy rates (hence why Youden-maximised thresholds were more common for PGSI harm classification). This demonstrates the need to carefully consider the metric(s) by which limits are selected and how the proportion of positive and negative cases in a sample may impact outcomes.

All threshold values for classifying PGSI and GHM harm we identified were higher than those observed in previous attempts to derive Australian guidelines [11, 19]. The outcomes suggest our proposed limits will result in similar or superior classification accuracy. Our limits provide a more even balance of sensitivity and specificity, whereas Dowling and colleagues' [19] wagering-specific limits have elevated sensitivity rates and notably lower specificity rates resulting from lower threshold values. Consequentially, when the two sets of limits are applied at a population level there will be a more meaningful impact on overall accuracy rates where the rates of harm are lower than reported here. For example, in Dowling and colleagues' [19] population prevalence survey sample, only 8.34% and 13.49% of horse/dog racing bettors and sports/other events bettors reported harm on the PGSI, compared to slightly over 50% of the unweighted

Table 5 Rates of customers above each new optimal limit in the six months before and after participation: overall and by harm status

Limit	Six months pre-survey					Six months post-survey				
	Overall	PGSI harm		GHM harm		Overall	PGSI harm		GHM harm	
		Harm	No harm	Harm	No harm		Harm	No harm	Harm	No harm
Betting days per month										
PGSI limit: >6 days	521 (49%)	316 (54%)	205 (43%)	190 (56%)	331 (46%)	475 (45%)	275 (47%)	200 (36%)	171 (43%)	304 (35%)
GHM limit: >9 days	383 (36%)	237 (41%)	146 (31%)	141 (42%)	242 (34%)	342 (32%)	209 (36%)	133 (22%)	124 (29%)	218 (23%)
Spend per month										
PGSI limit: >\$250	609 (58%)	396 (68%)	213 (45%)	252 (75%)	357 (50%)	549 (52%)	352 (61%)	197 (33%)	224 (55%)	325 (35%)
GHM limit: >\$2,100	314 (30%)	223 (38%)	91 (19%)	153 (45%)	161 (22%)	256 (24%)	179 (31%)	77 (5.7%)	116 (18%)	140 (6.5%)
% of monthly income spent per month										
PGSI limit: >6%	494 (47%)	341 (59%)	153 (32%)	228 (68%)	266 (37%)	439 (42%)	293 (50%)	146 (21%)	193 (44%)	246 (23%)
GHM limit: >9%	434 (41%)	307 (53%)	127 (27%)	213 (63%)	221 (31%)	383 (36%)	262 (45%)	121 (15%)	174 (38%)	209 (17%)
Number of different activities bet on										
PGSI limit: >4	470 (45%)	306 (53%)	164 (35%)	191 (57%)	279 (39%)	422 (40%)	279 (48%)	143 (25%)	175 (47%)	247 (28%)
GHM limit: >5	366 (35%)	246 (42%)	120 (25%)	150 (45%)	216 (30%)	317 (30%)	215 (37%)	102 (17%)	138 (35%)	179 (19%)
Monthly deposit frequency										
PGSI limit: >4	473 (45%)	340 (59%)	133 (28%)	224 (66%)	249 (35%)	390 (37%)	276 (48%)	114 (24%)	181 (54%)	209 (29%)
GHM limit: >14	297 (28%)	226 (39%)	71 (15%)	153 (45%)	144 (20%)	222 (21%)	166 (29%)	56 (12%)	107 (32%)	115 (16%)
Monthly deposit amount										
PGSI limit: >\$80	623 (59%)	419 (72%)	204 (43%)	266 (79%)	357 (50%)	514 (49%)	344 (59%)	170 (36%)	215 (64%)	299 (42%)
GHM limit: >\$900	312 (30%)	226 (39%)	86 (18%)	155 (46%)	157 (22%)	209 (20%)	149 (26%)	60 (13%)	94 (28%)	115 (16%)
% monthly income deposited per month										
PGSI limit: >2%	495 (47%)	354 (61%)	141 (30%)	233 (69%)	262 (36%)	393 (37%)	275 (47%)	118 (25%)	182 (54%)	211 (29%)
GHM limit: >10%	313 (30%)	231 (40%)	82 (17%)	167 (50%)	146 (20%)	213 (20%)	157 (27%)	56 (12%)	102 (30%)	111 (15%)
Active gambling accounts (12 months pre-survey)										
PGSI limit: >2	589 (56%)	369 (64%)	220 (46%)	229 (68%)	360 (50%)	589 (56%)	369 (64%)	220 (46%)	229 (68%)	360 (50%)
GHM limit: > 3	321 (30%)	208 (36%)	113 (24%)	134 (40%)	187 (26%)	321 (30%)	208 (36%)	113 (24%)	134 (40%)	187 (26%)

Values = frequency count (%). All figures are based on the 1,055 people who reported their income and thus their betting relative to *all* limits was known

sample here. Thus, having lower limits that emphasise sensitivity rates will lead to marginal improvements in accuracy by increasing the small number of true positives when harmed individuals comprise only a small percentage of the total sample, but this will be offset by substantially greater reductions in overall accuracy as a result of large increases in the number of false negatives (i.e., the *precision* or *positive predictive value* will become very small). Whether it is better to overemphasise sensitivity at the expense of specificity in practice remains unclear,

however the credibility of lower-risk guidelines is likely to be questioned if most of those gambling above them do not experience harm. Further research is needed to confirm which set of limits are more effective at differentiating lower from higher-risk gambling.

Overall, our limits are more similar to those identified by Louderback et al. [12] for online gambling who similarly used a methodology involving account-based data combined with self-reported harms. This suggests that limits derived from objective behavioural account data

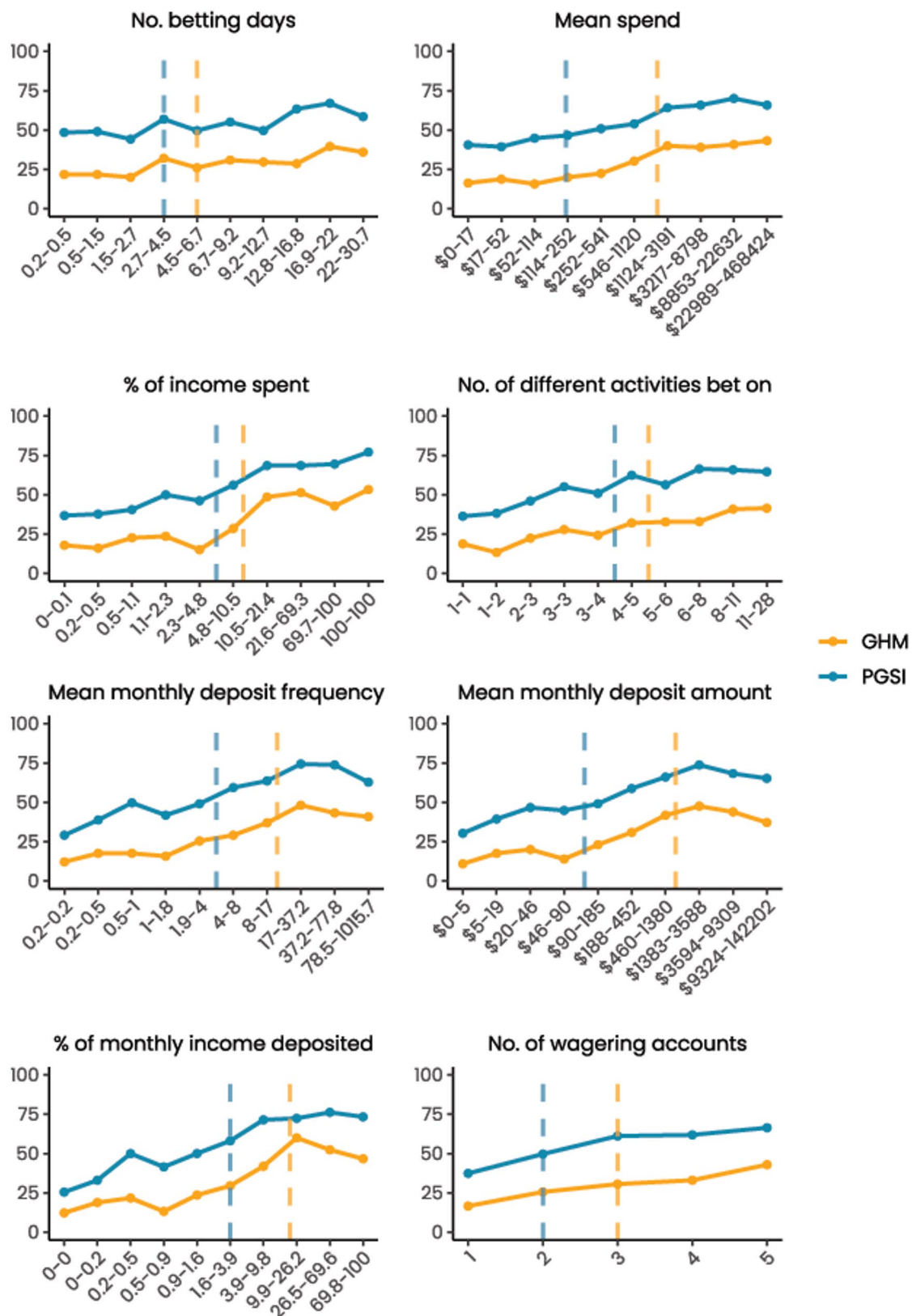


Fig. 3 Risk curves for the eight behavioural indicators. Risk curves for showing the proportions of participants within each decile for each of the eight behavioural indicators who reported harm according to the PGSI (blue) and GHM (orange). Dashed vertical lines represent the approximate location of the limits established here for each behavioural indicator (PGSI limit = blue; GHM limit = orange)

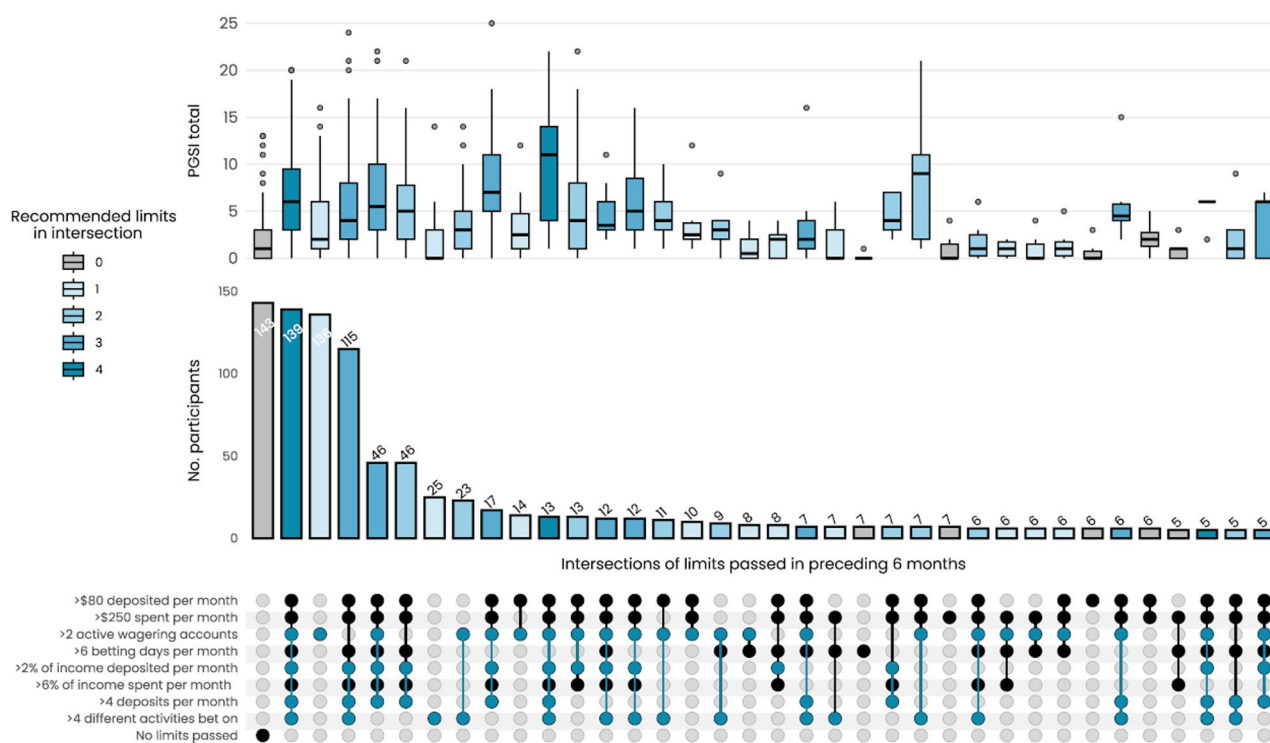


Fig. 4 Number of participants in limit passing intersections and their associated PGSI scores. Upset plot presenting all intersections of limit passing in the six months preceding survey participation – i.e., every possible combination of passing or not passing all eight PGSI-related limits. The bottom panel displays the intersections, with blue dots representing one of our four recommended limits (see Discussion section), and black dots representing those omitted from our recommendations. The middle plot displays the unweighted count of participants in each intersection. The upper plot shows boxplots based on the unweighted total PGSI scores for each intersection. Only the 1,055 people who reported their income were used to create this figure

are typically larger than those derived from self-reported behavioural data and is consistent with evidence of underreporting gambling frequency and spend in surveys [14, 45, 46].

As an exploratory exercise, we replicated our limit selection analyses using only a subsample of young participants aged 25 and under. Younger age has been consistently associated with an elevated risk of gambling problems [22, 35]. Our analyses indicated that lower limit values were appropriate for young people on all behavioural indicators other than number of accounts, suggesting they are more likely to experience harm at lower intensities. Whilst it might be assumed that the elevated base rates of harm according to the PGSI and GHM in this subsample resulted in limits that maximised sensitivity and were therefore lower, limits identified by all three approaches (including specificity-maximisation) were lower for this subsample.

Comparing the clinical and practical utility of each limit

Our analyses indicate that the eight behavioural indicators and the limits we identified for them varied substantially in their ability to differentiate lower from higher risk gambling. Starting with the limit with the least value in this respect, we found that frequency, operationalised

as the number of betting days per month, was not a strong indicator of risk. This was the only behavioural indicator for which the AUC value was less than 0.6 (our preregistered minimum level of classification ability). Further, when all limits were included in a regression model simultaneously, average betting frequencies above these values was associated with a reduced likelihood of reporting harm. The risk curve for betting frequency further illustrates its limited value in this context, showing the “flattest” trendline of any behavioural indicator—as betting days per month increased, there was little rise in the proportion of participants who reported harm (see Fig. 3). These findings are consistent with prior research, although they run contrary to the inclusion in betting frequency as a standard lower-risk guidelines. Dowling et al. [19] found that exceeding their limit for gambling frequency for sports/other events betting was not predictive of harm once other limits were included in their logistic regression model. Louderback and colleagues [12] found that while surpassing their gambling frequency limit (11 days per month) was associated with a positive screen on the Brief Biosocial Gambling Screen [47], it was associated with a reduction in the other risk indicators examined, including account closure and voluntary self-limiting. It is unknown why exceeding the gambling

frequency thresholds identified here was associated with a reduced likelihood of reporting harm, although it may be that higher-risk gambling in this sample was represented by more variable gambling patterns in response to available funds. In a separate analysis of the data presented here, increased betting and depositing around certain days of the week were observed, appearing to reflect a response to common pay days and sporting events in Australia [48].

The four “standard” behavioural indicators most similar to those used in the Canadian [7] and Australian [11] guidelines—betting frequency, amount spent, percentage of income spent, and breadth of gambling activity (defined as number of different sports & races bet on here)—mostly performed worse than the wagering-specific indicators examined (i.e., number of deposits, deposit amount, percentage of monthly income deposited per month, and number of wagering accounts), highlighting the importance of considering activity-specific behavioural indicators for lower-risk guidelines. For example, when predicting PGSI harm, none of the limits for the four standard behavioural indicators were significantly predictive of harm after accounting for the impact of all limits simultaneously. Only gambling above the limit for percentage of monthly income spent per month remained significantly predictive of GHM harm. We argue below, that ‘spend’ is a more difficult variable for consumers and operators to define and calculate, and hence this variable, while having some predictive utility, is not helpful to include as a lower-risk guideline.

Exceeding limits for deposit frequency, the percentage of monthly income deposited per month, and the number of active wagering accounts remained significantly and strongly predictive of PGSI and GHM harm in models including all limits simultaneously. The importance of these behavioural indicators is further displayed in Fig. 3 through the slope of each line. These findings have important consequences for understanding risk in the context of online wagering. First, increased expenditure on gambling relative to one’s household income is a substantially stronger indicator risk than absolute expenditure. Second, depositing behaviours appear strongly linked to risk and may reflect loss chasing behaviour, gambling beyond intended amounts, or loss of control [49]. Third, the number of active wagering accounts someone currently holds may be an important risk factor in this population. This indicator may be both a cause and a consequence of gambling harm. For instance, individuals experiencing harm might switch between platforms in search of better outcomes or to avoid detection by operators monitoring risky behaviour. Additionally, being registered on multiple sites may increase their exposure to advertisements and promotional offers that contribute to harmful gambling.

Strengths and limitations of the current study

Our study possesses several strengths. It is the first to focus on developing limits specific to online wagering, including limits for standard and activity-specific behavioural indicators validated against harm according to the PGSI and the GHM, and one of only two studies to have used objective behavioural account data to derive lower-risk guidelines [12]. We used bootstrapping to select optimal threshold values for eight different behavioural indicators using three separate approaches; employed weighted logistic regression models to validate the predictive ability of these limits after accounting for other, non-wagering gambling and demographic factors; and present all outcomes transparently here and/or in our online Analysis Document containing extensive supplemental outcomes. We also compare the performance of our limits with those most similar developed by Dowling et al. [19] and demonstrate the need to potentially set lower lower-risk threshold values for younger populations. Lastly, our study uses some of the most recently available data (2023–2024) to derive these limits. For example, despite all being published in 2021, Louderback and colleagues [12] note that their samples were “more than nine years old” and the studies by Dowling et al. [11, 19] used data from surveys collected between 2011 and 2015. Further, Currie et al. [1] used data from a longitudinal survey that ran from 2006 to 2011. Whilst all these studies have provided strong insights into the selection and value of lower-risk gambling guidelines, using more recent data to derive accurate limits may be particularly important in the context of the ever-changing online gambling landscape.

There are several limitations to our study. In a separate analysis of the dataset used for this study, Heirene et al. [39] found the sample of customers who completed the survey were older and more frequent bettors relative to all of those invited, meaning our limits may be skewed towards those more involved in online gambling. Further, whilst using objective behavioural account data to derive limits avoids biases in recall or self-preservation when reporting expenditure, it only captures gambling with one site. We attempted to account for this by including the self-reported number of wagering accounts somebody held and their other recent gambling as covariates in statistical models, though these are unlikely to fully capture the variety and extent of gambling outside of the site participants were recruited from. One solution to overcoming the limitations of both self-report and single-operator data for developing guidelines may be to use open banking data. This can provide objective spending data across all operators that an individual gambles with, as well as comparisons to income and other expenses [50, 51].

Practical applications and recommended future research

Like Louderback and colleagues [12], we call for further evaluations of our newly established limits for online wagering in other samples and against other indicators of risk (e.g., self-exclusion, setting, & changing/setting deposit limits). One notable caution is that online wagering behaviour is highly dynamic and influenced by government policies, available gambling formats and operator actions such as promotions. Research shows that changes to how products are provided are likely to influence gambling behaviour; for example, slowing reel spins for EGMs was trialled in an attempt to reduce gambling losses but resulted in customers wagering more per spin [52]. Thus, if policies were introduced such as to cap deposit limits, this would likely change this behaviour and the predictive value of depositing in identifying risk. Nonetheless, we proffer recommendations for how our outcomes could be used to inform harm reduction efforts and how future research can support these efforts.

First, comparing the optimal limits identified for PGSI and GHM harm revealed that the latter resulted in substantially higher threshold values, likely due to a higher perceived level of harm being required for participants to endorse the items on this measure. We initially suggest the limits for PGSI harm are likely to be of greater value than those for GHM harm if wanting to ensure adequate sensitivity levels for lower-level harms.

Second, we recommend focusing on depositing as a key behavioural indicator as this is an unambiguous transaction with little room for misinterpretation and our results show that this is a significant indicator of risk. Conversely, in the context of online wagering, 'spend' is commonly interpreted in different ways [53]. For example, some consumers may consider spend the amount lost (net outcome), the amount deposited, or the total amount staked (as used here)⁷. Deposits are arguably easier for consumers to mentally track than spend. Online wagering increases the ability to track these spend within accounts, depending on how this is defined, however, deposits have the additional benefit of being tracked independently and across operators by consumers through payment providers and contrasted with any withdrawals. It has been recommended that banks include gambling as a specific category of spend in summaries for customers to assist customers with tracking their spend across other expenses such as utilities, groceries, and entertainment [50].

Third, our findings may provide an empirical basis for income-relative limit setting, whether voluntary or government mandated. Concerns and uncertainty regarding

the specific amount that one should set as their deposit limit was one of the key barriers to engagement with this tool identified in a study of Australian online wagering customers [54]. The threshold value of 2% of monthly household income deposited per month provides a guide for consumers who wish to set limits on their gambling that are specific to their financial situation. This value can also guide operators who are required to conduct well-being and affordability checks, as well as financial institutions who may consider gambling as part of a credit assessment. Relative expenditure is likely more useful in identifying risky gambling than specific values proposed in the UK and Netherlands affordability policies. Income-relative limits may also be perceived as more appropriate by consumers. However, operationally, all customers would have to provide salary data to enable this limit to be checked, thus still necessitating value-based levels as a cut-point for further assessments.

Our findings indicate that the relationship between single behavioural indicators and harm is not one of a steep, linear trendline, consistent with previous research on behavioural indicators for gambling risk [49]. Yet, the odds-ratios presented here (Fig. 2) indicate that exceeding any of the recommended levels increases the risk of experiencing gambling harms; therefore, our findings do not suggest that breaching a single limit is without risk. Rather, exceeding multiple limits compounds the risk exponentially, and therefore warrants prompt action by both the individual and operators if observed. Conversely, a notable minority of individuals who gamble below limits still report harm (hence why we have opted to use the term "*lower-risk guidelines*" rather than "*low-risk guidelines*"), and thus stakeholders should not consider sub-limit gambling risk-free or 'safe'.

The perceived value and impact of our recommended limits remain unknown. If consumers do not find the limits developed here to be credible then their value as harm reduction measures may be low. One recent study in Finland used surveys to assess the perceived feasibility and acceptability of the Canadian low-risk gambling guidelines [10], finding broad support for their overall value as a harm reduction tool and for the specific threshold values set for the limits. Replicating this work in an Australian sample with the limits developed here will be an important step in evaluating their harm-reduction potential as well as the practical utility in terms of how to encourage gambling customers to adhere to these.

Considering all our findings, we tentatively propose four lower-risk wagering guidelines which we have graphically presented in Fig. 5. The limits for percentage of monthly income deposited per month, number of deposits per month, and number of active wagering accounts were the most consistent and empirically supported predictors of gambling harm, making them

⁷ This was evident in Currie et al.'s (2017) study, wherein the authors needed to recode responses to questions regarding gambling spend in which participants that indicated that they had won money, changing them to "0".



Fig. 5 Recommended lower risk guidelines for online sports & race betting Australia

obvious candidates for inclusion in our guidelines. The limit for the number of sports and races bet on was included based evidence of increasingly greater risk of harm with each limit passed—and thus value in providing consumers with a greater number of limits—and the exclusion of all other behavioural indicators for practical or statistical reasons. As discussed, the two limits relating to “spend” may be misinterpreted by consumers and so were excluded. The limit relating to absolute amount deposited was excluded to avoid potentially conflicting with the limit for percentage of income deposited, which provides more personally meaningful guidance. Finally, the betting frequency limit was removed based on evidence of limited effectiveness as a lower-risk indicator. Our exploratory analyses indicate that almost all harmed participants in our sample passed one or more of our four recommended limits, suggesting that this selection does not fail to capture a large at-risk group by omitting the other four limits.

Displaying the guidelines in an infographic-style format in Fig. 5 allows for them to be presented alongside supportive information (e.g., conversions from annual income to maximum monthly deposit values) and key

considerations (i.e., the need for some groups to consider setting lower limits). We adapted portions of the text in Fig. 5 from the Canadian Low Risk Gambling Guidelines documentation (see: <https://gamblingguidelines.ca/resource/lower-risk-gambling-guidelines/>) to ensure consistency in framing. We adopted the slogan “The 2-2-4-4 Rule” with the aim of using the numerical pattern as a mnemonic (i.e., a memory aid). Mnemonics have shown to significantly enhance learning and recall, particularly when conveying abstract or complex information [55]. Mnemonic devices have been effectively used in both public health campaigns—such as Australia’s “Slip, Slop, Slap, Slide” message outlining behavioural recommendations for sun protection—and commercial marketing contexts [56–59]. By harnessing this approach, we aimed to make the guidelines more memorable, ultimately increasing their usefulness for the target end user.

Conclusions

This study has established lower-risk guidelines for online sports and race betting in Australia using an extensive series of preregistered and exploratory analyses of objective behavioural account data with matched survey

responses. These guidelines provide specific recommendations for those who choose to gamble that can guide them to do so in a lower-risk manner. The proposed guidelines provide advice that is relative to one's financial situation making them more targeted and actionable than the common instruction to "Gamble responsibly" or warnings such as "What are you really gambling with?" [6]. Our recommended guidelines are [1] make no more than four deposits per month, [2] deposit no more than 2% of your pre-tax household income per month, [3] bet with no more than two different sites or apps, and [4] stick to betting on four or fewer different types of sports or races. Exceeding any of these guidelines corresponds to an increase in the risk of experiencing gambling harms and exceeding multiple results in an exponential risk increase. Further research is required to refine and adapt these to specific cohorts (e.g., younger adults), assess the perceived acceptance of the guidelines among consumers, and develop implementation practices to encourage adherence.

Author contributions

Robert M. Heirene: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Visualization, Writing—original draft, and Writing—review & editing. Dilushi Chandrakumar: Conceptualization, Investigation, and Writing—review & editing. Gary Fahey: Conceptualization and Writing—review & editing. Eunice L. Y. Huynh: Visualization and Writing—review & editing. Sally M. Gainsbury: Conceptualization, Funding acquisition, Investigation, Supervision, and Writing—review & editing.

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Data availability

Data used for this study are commercially sensitive and cannot be shared beyond the research team. Instead, all analysis code and outcomes extending beyond those presented in this manuscript are included in our Analysis Document available here: <https://robheirene.quarto.pub/lower-risk-guidelines-for-online-wagering-analysis-document/>.

Declarations

Ethics approval and consent to participate

Ethics approval was obtained from the University of Sydney Human Ethics Committee [ID: 2023/029] and informed consent was obtained from participants at the beginning of the data collection survey. The online operator provided account data for their customers, who agree for their de-identified data to be shared with approved third parties during registration. Individuals who took part in the survey were aware that we would collect their account data before they provided consent.

Consent for publication

Not applicable.

Competing interests

RH has worked on a project funded by Responsible Wagering Australia (a representative body of Australian online wagering operators; University of Sydney, 2019–2021) and as an independent, sub-contracted statistical consultant for PRET Solutions Inc on a commissioned project (funded by the Australian Casino operator Crown; 2023). RH has received funding from the International Center for Responsible Gaming (ICRG) and the Brain and Mind Centre, University of Sydney. DC has worked on projects funded through the University of Sydney from West HQ and the ICRG. GF and ELYH have worked on projects funded through the ICRG, awarded to SG and RH. SG has received direct and indirect funding since 2020 through the University of Sydney from Australian Leisure and Hospitality Group Pty Ltd, Entain Australia, Sportsbet, NSW Office of Responsible Gambling, West HQ, Brain and Mind Centre, Cambridge Health Alliance, Aristocrat Leisure Limited, and ICRG. SG has received consulting funds for providing subject matter expertise for New Zealand Crown Counsel, NSW Liquor & Gaming Authority, New Zealand Gaming Machine Association, Singapore Ministry of Health, Betcloud, NZ BlueCloud, UK Behavioural Insights Team, KPMG, QBE, Coms Systems Limited, Advance Gaming (NZ) Limited, GambleAware, Star Entertainment, GREO, Senet, and Norths Collective. SG has received honorarium and/or travel costs for presentations for Cyprus National Betting Authority, Asian Racing Federation, Leagues Club Australia, Australian Cricketers Association, Star Entertainment, CAMH, Behavioural Insights Team, National Council on Problem Gambling, GambleAware, GREO, Informa, and Washington State Council, European Lotteries Association. SG holds unpaid appointments as an invited member on the NSW Independent Panel on Gambling Reform, is the Pillar Champion for Technology and Environment for the QLD Responsible Gambling Advisory Committee, is a board member for the Asian Racing Federation Council on Anti-Illegal Betting and Related Financial Crime. SG receives an honorarium for her role as Co-Editor-in-Chief for International Gambling Studies from Taylor & Francis.

Author details

¹School of Psychology, Brain & Mind Centre, Science Faculty, University of Sydney, Sydney, Australia

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